



Developing a novel antibacterial coating system for marine corrosion and fouling control

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ABSTRACT

Microbial-influenced corrosion and marine biofouling have become a thorny problem restricting the effective long-term operation of marine engineering. In this study, a novel antibacterial coating system for marine corrosion and fouling control was developed to integrate and improve the antibacterial performance. The coating system was composed of an epoxy primer and a low surface energy organosilicon/polyurethane topcoat, while quaternary ammonium salt (QAS) was added as an antimicrobial agent. The primer provided excellent corrosion protection. The antifouling performance was significantly improved by the combined effect of low surface energy and antibacterial agents, resulting in an extended service life and enhanced environmental sustainability of the coating system. Furthermore, the satisfactory results of the above coating system were confirmed by 8 weeks of hanging tests in the ocean, which exhibited potential application prospects in marine engineering in the future.

1. Introduction

In the field of marine engineering, corrosion of metal materials and biological pollution are thorny problems that restrict the long service life of marine equipment. The ocean has a highly complex natural environment. The attachment and reproduction of marine fouling organisms on vessels, pipelines, and other marine equipment surfaces can significantly affect the regular operation of marine instruments and equipment, shorten their lifespans, and cause safety hazards (Xu et al., 2015; Hmamou et al., 2013; Li et al., 2015). In particular, the corrosion of metallic materials might be accelerated by biological fouling of coated metal substrates, known as microbial-influenced corrosion (MIC) (Lou et al., 2021). MIC is reported to account for 20 % of all corrosion failures and causes economic losses of \$30 to \$50 billion per year (Chugh et al., 2020; Hou et al., 2017). Surface coating of ships and marine equipment is one of the most important methods to prevent corrosion and marine biological pollution. However, to the best of our knowledge, the current research mainly focuses on a single anti-corrosion coating or anti-fouling coating, and little has been heard of research on anti-corrosion and anti-fouling supporting coating systems. Therefore, to combat the synergistic effect of biological pollution and corrosion on metal materials, there is an urgent need to construct a matching coating system with dual functions of antifouling and anti-corrosion.

Marine paint is often divided into primer and topcoat in marine engineering practice. Primer is the first coating layer on the substrate, which should have good adhesion, mechanical strength, and corrosion resistance. At the same time, due to the specificity of the marine environment, it is necessary to have a certain degree of antimicrobial properties (Liu et al., 2020). The topcoat should have excellent physical and antifouling properties as a coating in direct contact with the marine environment. Therefore, selecting antimicrobial agents with good adaptability, stable presence, and perfect antimicrobial effect is the core of this work.

There are three main types of anti-bacterial agents for coatings: natural anti-bacterial agents, organic anti-bacterial agents, and inorganic anti-bacterial agents (Zhang et al., 2018a). Inorganic anti-bacterial agents include silver antibacterial agents, zinc oxide, titanium dioxide, etc. (Xu et al., 2017; Selim et al., 2015). Organic antibacterial agents include quaternary ammonium salt, isothiazolinone, etc. (Sun et al., 2022; Wang et al., 2020; Ferreira et al., 2020; Liu et al., 2022; Muñoz-Bonilla and Fernández-García, 2012). Natural antibacterial agents refer to substances with anti-bacterial activity extracted directly from animals, plants, or microorganisms, mainly including chitosan, protamine, allicin, sorbic acid, etc. (Ma et al., 2017; Mitra et al., 2016; Nigmatullin et al., 2009; Qian et al., 2010). Oktay and Kayaman-Apohan (2013) added nano-silver to the polysiloxane

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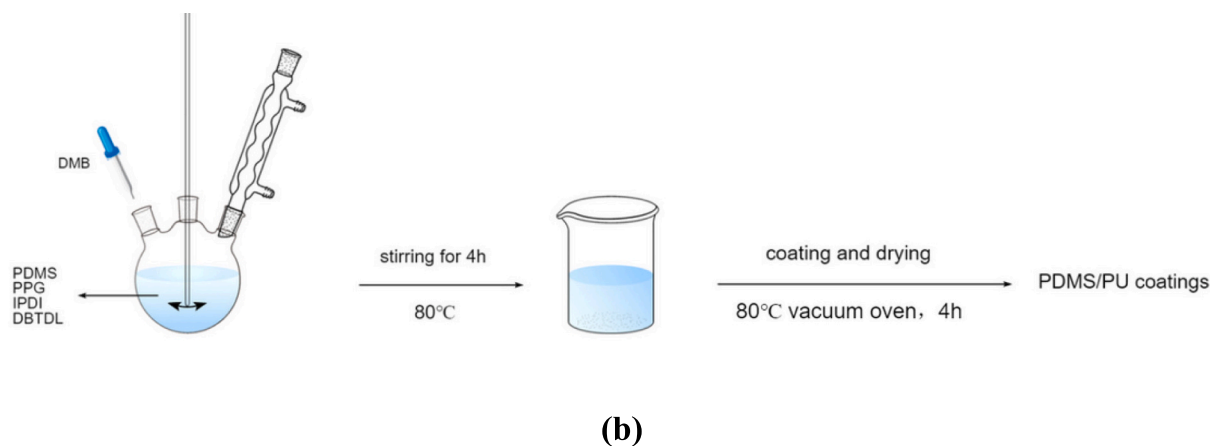
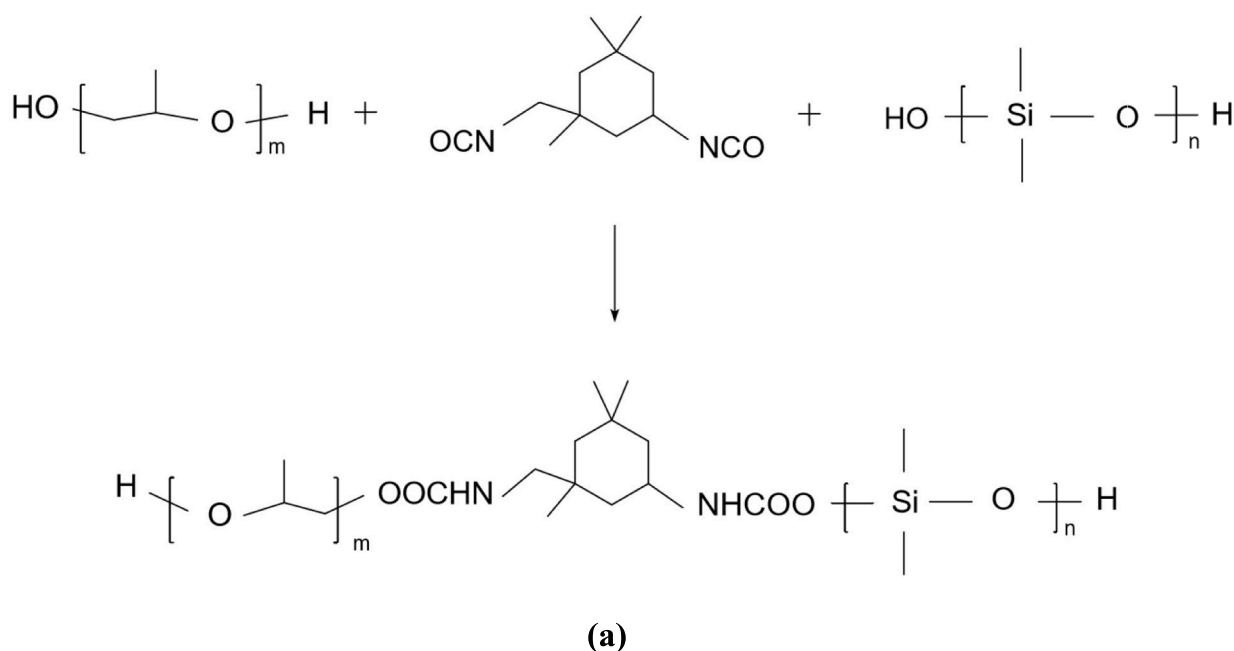


Fig. 1. (a)Synthetic route and (b) reaction flow chart of PDMS/PU random copolymer coatings.

coatings. They found that the bactericidal rates against *Escherichia coli* and *Staphylococcus aureus* all reached >99 % without affecting the static water contact angles of the coatings. Qu et al (Yuanyuan and Hao, 2015) added nano titanium dioxide powder into the silicon-modified acrylic resin. They found that the anti-bacterial properties of the coatings were greatly improved, and the surface roughness and the water contact angles of the coatings were improved simultaneously. Sauvet et al. (2000) added quaternary ammonium salts to the silicone resins. The obtained coating has bactericidal activity against *E. coli* and *Aeromonas hydrophila*. The activity remained unchanged after 66 days of immersion in water. Researchers (Wispongpan, 2003) found that phloroglucinol extracted from brown algae has an excellent anti-bacterial effect against *Bacillus subtilis* and *S. aureus*, which is expected to be applied to the coating as a natural anti-bacterial agent.

Quaternary ammonium salt (QAS) was one of the most studied organic anti-bacterial agents, with several advantages over other anti-bacterial agents, including excellent cell membrane penetration properties, lower toxicity, less skin irritation, and better environmental stability. Its anti-bacterial mechanism is mainly combined with anions on the cell membrane surface of bacteria, which destroys the synthesis system of proteins and cell membranes, thereby inhibiting the reproduction of bacteria (Xie et al., 2011; He et al., 2015). Poverenov and

Klein (2018) recently published a review that describes the significant anti-bacterial activity of QACs on different surfaces, including synthetic and natural polymers, glass, and metal surfaces. This survey shows that the covalent linkage of QACs might establish a new platform for developing potential, safe, and sustainable anti-bacterial materials. Moreover, QAS combined with additional active agents might result in new materials with multifaceted activity (Sundrarajan and Rukmani, 2013; Dong et al., 2018).

In addition, the selection of topcoats, tin-free self-polishing anti-fouling coatings, conductive anti-fouling coatings, biological and bionic anti-fouling coatings, low surface energy coatings, and others have been applied in practice (Callow and Callow, 2011; Selim et al., 2017). Among them, the low surface energy anti-fouling coatings achieve anti-fouling by utilizing the low surface energy and the shearing action of water flow (Lejars et al., 2012). There is no release of any toxic substances in the process of action. Therefore, low surface energy coatings have gradually become the research focus in related fields.

Low surface energy coatings are generally divided into organic silicon and organic fluorine, among which organic silicon series polymers are widely used considering cost, toxicity, and other factors (Buskens et al., 2012; Ciriminna et al., 2015; Król et al., 2017). Low surface energy coatings have low surface energy due to the high flexibility of molecular

chains (Duong et al., 2015; Lei et al., 2018) and poor mechanical properties. Blending or copolymerizing with epoxy resins, polyurethane resins, phenol-formaldehyde resins, and others can effectively improve the mechanical properties of silicone resin coatings (Hu et al., 2020; Lejars et al., 2020). Sun et al. (2019) copolymerized polyorganosiloxane with bisphenol A epoxy resin and found that the adhesive strength and tensile strength of the coatings were significantly improved. It also had an excellent anti-fouling performance. Zhang et al. (2018b), Sommer et al. (2010) and Naghash et al. (2013) prepared silicone resins with hydroxyl-terminated respectively and copolymerized them with isocyanate. It was found that the copolymer coatings had excellent mechanical properties and low surface energy characteristics simultaneously. Chen et al. (2008) added nano-SiO₂ powder into the organosilicon-modified acrylate resin and found that the coatings have quality anti-fouling and mechanical properties through characterization. The modification of silicone resins might affect some properties of the coatings, so balancing the anti-fouling performance and mechanical properties of the modified silicone coatings is one of the focuses of current research.

Therefore, in this paper, QAS is used as the antimicrobial agent, which is compounded with epoxy resin and low surface energy organosilicon/polyurethane, and the optimal amount of QAS and the optimal formulation of low surface energy organosilicon/polyurethane are determined by the comprehensive mechanical property test, electrochemical test, antimicrobial property test, and various macro and micro tests. The high-performance antimicrobial anti-corrosion coating supporting system is prepared, which provides a practical solution for the long-term development strategy of marine equipment.

2. Experimental section

2.1. Materials

Epoxy resin and curing agent (HY591) were purchased from Huaou Chemical Trading Co. Ltd. (Wuxi, China). The sodium chloride (NaCl), dimethyl benzene, methanol, and n-butyl alcohol (99.5 % purity) were purchased from Sinopharm Chemical Reagent Beijing Co. Ltd. (Beijing, China). Isophorone diisocyanate (mixture of isomers, IPDI, 99 %), polypropylene glycol (PPG, M = 400), hydroxy silicone oil (PDMS, M = 92.17), dibutyltin dilaurate (DBTDL, 95 %), hydrochloric acid solution (HCl, 1 M), sodium hydroxide solution (NaOH, 1 M) and sodium chloride (NaCl, AR), acetone (AR) were purchased from Macklin Company. Hexadecyl trimethyl ammonium bromide (CTAB, AR) was purchased as QAS from Tianjin Guangfu Fine Chemical Research Institute. Nutrient agar medium was purchased from Guangdong Huankai Microbial Sci. & Tech. CO., Ltd. The anti-bacterial property of synthesized materials was tested against *Staphylococcus aureus* (S. aureus, ATCC 6538) and *Escherichia coli* (E. coli, ATCC 25922) as Gram-positive and Gram-negative bacteria, respectively. The low-carbon steel was purchased from Bao Steel Co. Ltd. (Shanghai, China). The chemical composition (wt%) of the steel is listed in Table S1.

2.2. Preparation of the coatings

2.2.1. EP/QAS primer

The mixture of xylene and n-butyl alcohol (V1:V2 = 2:1) was used to dissolve the abovementioned QAS at 25 °C for half an hour. Epoxy resin and the QAS were put in low-carbon steel, in which the mass proportion of QAS was 1 wt%. After 1 h stirring, the curing agent was added into the can as epoxy resin/curing agent equal 4/1. The Q235 steels were subjected to abrasion with dry sandpaper of 400, 600, 800, and 1200 grade subsequently. Then, the samples were degreased by acetone, washed using anhydrous ethanol, and blown dry using a compressed air supply. The mixture was coated on the surface of the Q235 steel electrode with a bar coater. After curing at 25 °C for 7 days, the EP/QAS coating was obtained, which was called EP/QAS coating with 1 wt%. For

comparison, the different mass proportions of QAS (0 wt%, 3 wt%, and 5 wt%) were used to prepare the EP/QAS coatings according to the above procedure. The thickness of the coating films was controlled within 70 ± 5.5 μm.

2.2.2. PDMS/PU/QAS topcoat

A series of hydroxyl PDMS / PU copolymers with different ratios were synthesized by a one-step method (Zhang et al., 2015a). The synthesis route of low surface energy coatings is shown in Fig. 1. Firstly, hydroxy silicone oil, polypropylene glycol, and isophorone diisocyanate were added to the three-necked flask according to the set ratio, and then a small amount of dibutyltin dilaurate as the catalyst. Secondly, while stirring at 80 °C, an appropriate amount of dimethylbenzene was added to ensure that the viscosity of the liquid could be maintained within the proper range. After 4 h of stirring at 600 rpm, the final mixture was coated on the prepared low-carbon steel. Finally, the coatings were dried in a vacuum oven at 80 °C for 4 h to ensure the solvent would be volatilized entirely. The formulated coatings had a dry film thickness of 60 ± 5.5 μm.

Put the QAS powder in the mortar, ground it fully, weigh it according to the planned mass, and add it into the three-necked flask before the reaction starts. Other steps were the same as those in Fig. 1(b). Table S3 shows the compositions of the series of anti-fouling topcoats. The prepared coatings had a dry film thickness of around 90 μm.

2.3. Characterization

The morphology of the coatings was characterized using a field emission scanning electron microscope (FE-SEM, S4800). The composition of the low surface energy coatings was analyzed by Fourier Transform Infrared Spectroscopy (ATR-FTIR, NICOLET 6700) with a scanning range of 4000–500 cm⁻¹. The adhesion of the coatings was tested by a digital pull-off adhesion tester (BGD 500, Biaoge Da Precision Instrument (Guangzhou) Co. The flexibility grade of each coating was tested with a paint film flexibility tester (BGD 560, Sigma Precision Instruments (Guangzhou) Co., Ltd.), and the results were expressed in terms of the diameter (mm) of the shaft that did not cause the coating to crack twice significantly. The impact strength of each coating was tested with an impact tester (BGD 304, Sigma Precision Instruments (Guangzhou) Co. The results were expressed in terms of the height (cm) of the weight drop that did not result in significant cracking of the coatings on three occasions. The static water contact angle of the coatings was measured with a water contact angle tester (OCA 25, Data Physics Instruments GmbH) to characterize the surface energy of the coatings. Five parallel experiments were performed for each coating, and the average value was calculated as the final result.

2.4. Antibacterial experiments

The antibacterial properties of the coating were evaluated by using *S. aureus* (ATCC 6538) and *E. coli* (ATCC 25922). For the flat colony counting method, each sterilized coating was immersed in 200 μL bacterial suspension diluted with sterile saline solution to 10⁵ CFU/mL for 1 h. After being gently rinsed 3 times with sterile saline, the bacteria adhered to the coating surface were dispersed in 5 mL of a sterile salt solution. Subsequently, 100 μL of the dispersion was inoculated on a solid Luria Bertani (LB) agar plate and further cultured for 24 h at 37 °C. Finally, the morphology and state of the bacteria attached to the coating were observed by laser confocal microscopy. The number of *E. coli* and *S. aureus* colonies was counted, and the bacterial inhibition rate was calculated using the Eq. (1).

$$\text{Bacterial inhibition ratio (\%)} = (N_0 - N_x / N_0 \times 100\% \quad (1)$$

where N_0 is the number of bacterial colonies in the coating EP. N_x is the number of bacterial colonies in EP/QAS ($x\%$), which are $x = 1, 3$, and 5 ,

Table 1
Mechanical performance parameters of EP and EP/QAS coatings.

Sample	Impact strength (Kg. cm)	Flexibility (mm)	Adhesive strength (MPa)
EP	50	2	4.41
1wt%EP/QAS	>50	1	4.35
3wt%EP/QAS	>50	1	4.43
5wt%EP/QAS	>50	1	3.75

respectively.

2.5. Electrochemical measurements

The electrochemical measurements were conducted using the Autolab PGSTAT 302 N workstation to evaluate the corrosion behaviors of the coatings. The anti-corrosion properties of the coating were evaluated via electrochemical impedance spectroscopy (EIS) and potentiodynamic polarization (PDP). The electrochemical measurements were performed in a three-electrode system at room temperature in 3.5 wt% NaCl solution. A saturated calomel electrode (SCE) was used as the reference electrode, a platinum electrode was used as the counter electrode, and the studied sample was the working electrode. The electrochemical impedance was measured in the frequency range of 10^5 – 10^{-2} Hz with an AC range of 10 mV. A potentiodynamic polarization test was carried out from -0.25 to 0.25 V (vs. OCP) with a constant sweep rate of 1 mV/s. The corrosion potential (E_{corr}) and corrosion current density (I_{corr}) were obtained based on the Tafel extrapolation method. Before each EIS measurement, the open circuit potential (OCP) should reach a steady

status. All tests were repeated three times, and data analysis was performed using Nova2.1 software. The electrochemical impedance results were simulated according to suitable electrical equivalent circuits through ZSimpWin software.

2.6. Neutral salt spray test

The neutral salt spray test was performed according to GB/T177191 (Luo et al., 2018). The sample was placed in a salt spray chamber at 35 °C. The salt spray chamber was filled with a 5 wt% NaCl solution; the pH of the solution is about 6.5, and the spray volume per 80 cm² area should be (1–2.5) mL/h. After washing and drying with distilled water, the digital image of the sample was recorded periodically.

2.7. Ocean hanging test experiment

The plates were thoroughly polished and primed. After the primer was completely dry, the test coatings were applied. Finally, shallow sea hanging panels were obtained.

To test the antifouling properties of the coatings, the panels were immersed in the East China Sea at depths from 1 to 2 m in Xiamen, China, between March and April 2023. The panels were checked regularly to observe the progress of the fouling process.

3. Results and discussion

3.1. Mechanical properties and microscopic characterization of coatings

3.1.1. Characterization of the primer

Table 1 shows the impact strength and flexibility of the EP and its coatings contents obtained by the adhesion, impact strength, and bend

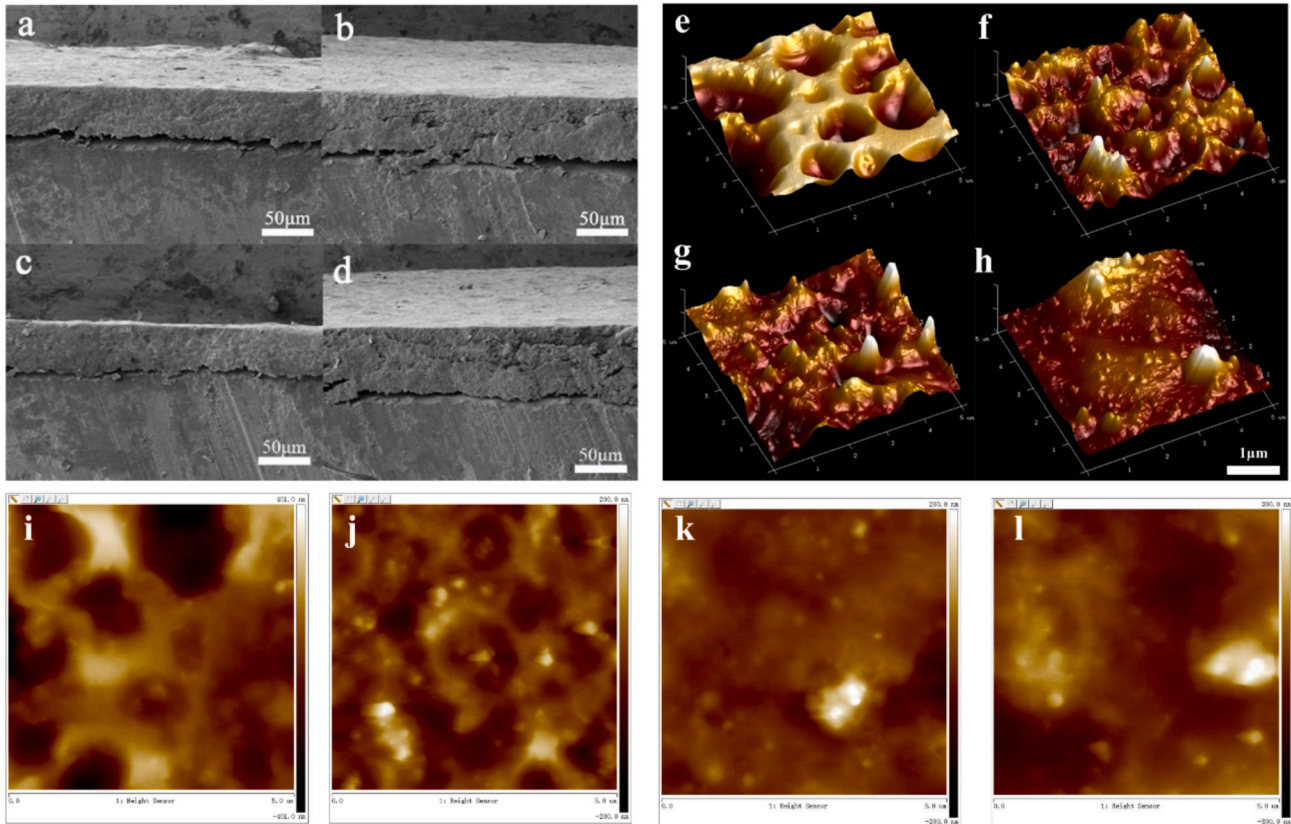


Fig. 2. SEM images of (a) EP, (b) EP/QAS (1 wt%), (c) EP/QAS (3 wt%), and (d) EP/QAS (5 wt%) coatings. Three-dimensional AFM topographies of (e) EP, (f) EP/QAS (1 wt%), (g) EP/QAS (3 wt%), and (h) EP/QAS (5 wt%) coatings. Two-dimensional morphology AFM image of (i) EP, (j) EP/QAS (1 wt%), (k) EP/QAS (3 wt%), and (l) EP/QAS (5 wt%) coatings.

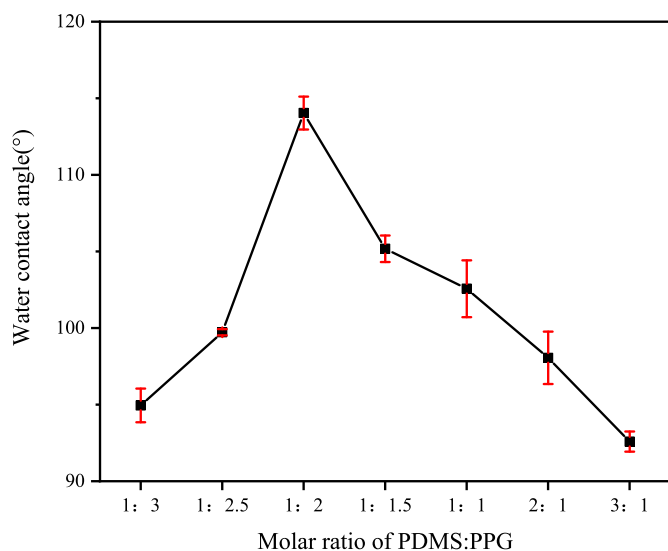


Fig. 3. Trend of static water contact angles with the change of PDMS and PPG ratio in the coating.

The molar ratio of PDMS and PPG is: (a) 1:3 (b) 1:2.5 (c) 1:2 (d) 1:1.5 (e) 1:1 (f) 2:1 (g) 3:1

tests. The results show that the impact resistance and flexibility of the coating are improved, and the adhesion strength of the coating is unchanged. However, when the doped contents of QAS further increase, the adhesion of the coating has declined. The reason might be that the excessive addition of QAS causes large pores in the coating. Therefore, compared with epoxy coatings, the mechanical properties of coatings are improved after the addition of QAS, but excessive addition will reduce the performance of coatings.

To further confirm the compatibility of QAS with the epoxy resin and the ability of the primer to bond to the substrate surface, the surface and cross-sectional morphology of the coatings and epoxy varnishes with different QAS contents were characterized by SEM. Fig. 2 shows the cross-sectional SEM images of the coatings with different QAS contents. The surface of the primer containing 3 wt% QAS has the smallest gap width, while the dense structure of the coating cross-section indicates good interfacial interactions between the epoxy resin and the quaternary

ammonium salt and between the matrix and the primer.

The AFM images of the EP/QAS coating are shown in Fig. 2 (e-h). The AFM images indicate that the surface of each coating shows fewer voids and cracks and has barriers to the penetration of corrosive ions. This observation confirms that the surface of the primer was intact after introducing the QAS into the epoxy coating. Fig. 2 (i-l) shows the AFM images of the two-dimensional morphology of the EP/QAS coating. The average roughness of the coating surface is reduced from 95 μm to about 25 μm after adding QAS, which shows that the coating surface is smoother, not easy to attach bacteria, and can have better antibacterial performance. In addition, the lower roughness value of EP/QAS (3 wt%) shows that it can protect the substrate more effectively than EP.

3.1.2. Characterization of the topcoat

Fig. 3, Figs. S5-S6 and Table S4 show the results of mechanical properties and static water contact angle tests of PDMS/polyurethane coatings with different ratios. The test results show that the best overall performance of the coatings was achieved when the molar ratio of PDMS to PPG was 1:2. The basic formulations of the coatings in the subsequent experiments met this requirement.

Fig. 4(a1) and (a2) show the FTIR spectra of the PDMS/PU coating

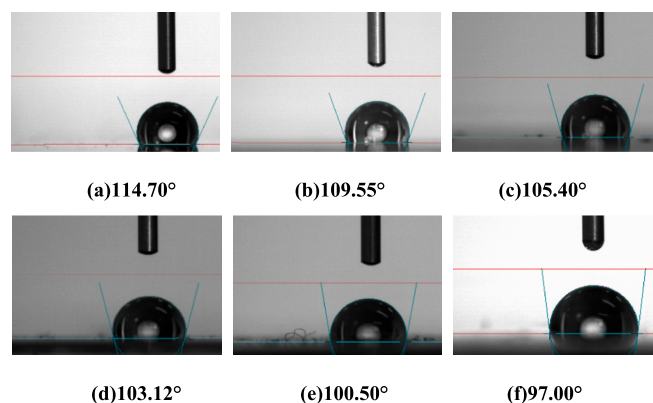


Fig. 5. Photos of static water contact angles of coatings with different content of QAS. The mass fraction of QAS is (a) 0, (b) 2 %, (c) 4 %, (d) 6 %, (e) 8 %, (f) 10 %.

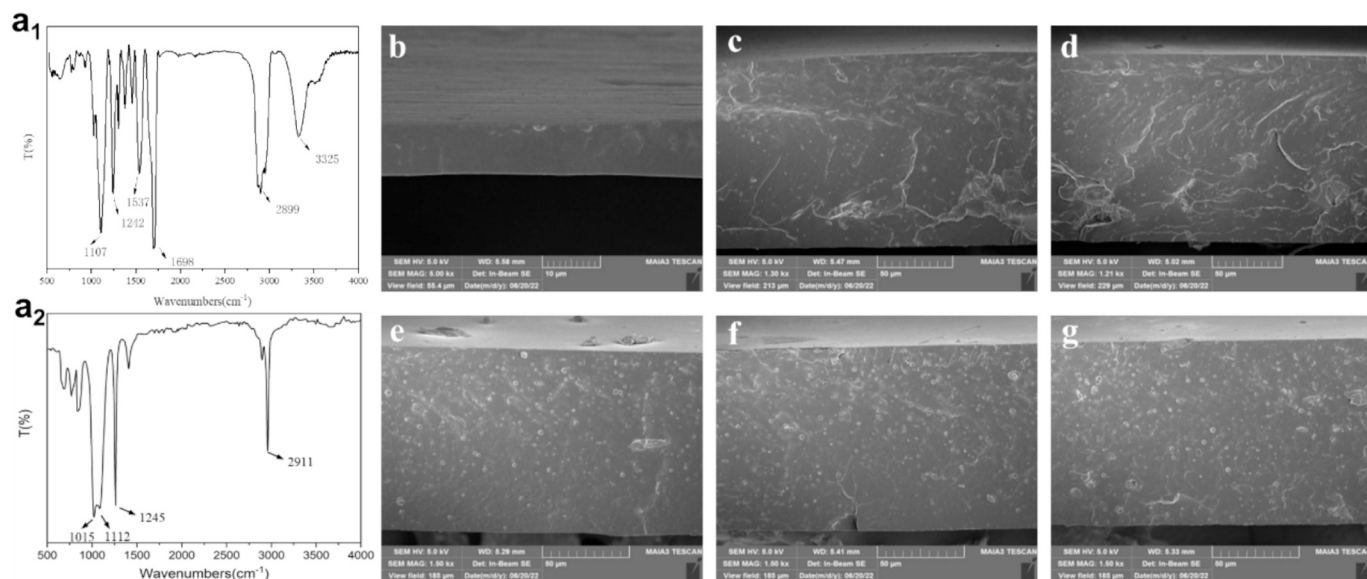


Fig. 4. FTIR spectra of (a1) the PDMS/PU coating and (a2) PDMS. SEM images of coatings with different QAS content. The mass fraction of QAS is (b) 0, (c) 2 %, (d) 4 %, (e) 6 %, (f) 8 %, (g) 10 %.

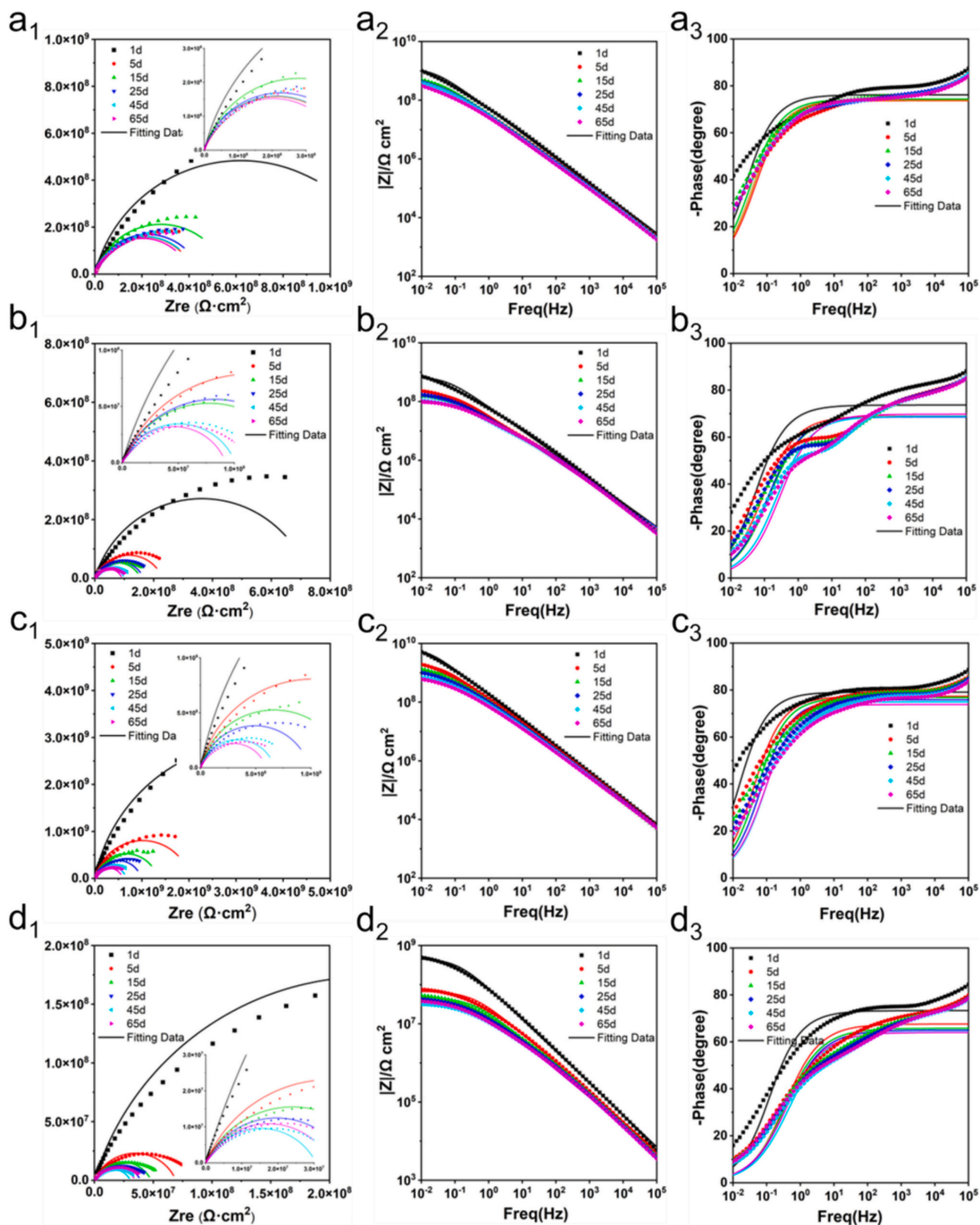


Fig. 6. EIS test of (a) EP, (b) EP/QAS (1 wt%), (c) EP/QAS (3 wt%), (d) EP/QAS (5 wt%) coatings.

and PDMS. The strong Si—O stretching absorption at 1107 cm^{-1} is the characteristic peak of the siloxane backbone, which can be seen in PDMS and PDMS/PU coating. The peaks at 3325 cm^{-1} (N—H stretching), 1698 cm^{-1} (C=O stretching) and 1537 cm^{-1} (C—N—H bending) are typical

chemical structures of polyurethane. The peak at 1242 cm^{-1} is mainly the peak of N—C—O stretching and Si—C bond. In addition, an aliphatic C—H elastic band is designated at 2899 cm^{-1} . Compared with the literature results, no obvious absorption peaks belonging to —NCO

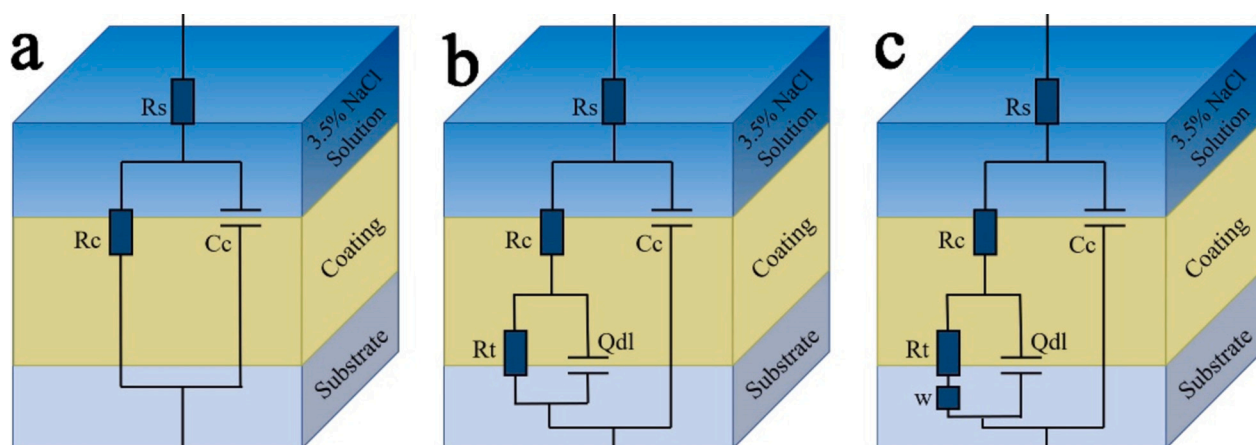


Fig. 7. The equivalent electrical circuits are employed to simulate the impedance data of the samples.

and-OH were observed around 2260 cm^{-1} and 3400 cm^{-1} , which proved that IPDI, PPG and PDMS had fully reacted. According to the FTIR test results, PDMS/polyurethane coating was successfully prepared.

The distributions of QAS in the coatings are shown in Fig. 4(b-g). As can be seen from the figure, QAS powder is evenly dispersed in the coating. At the same time, when the content of QAS reaches or exceeds 6 %, agglomeration will occur. At this time, white dots representing QAS will appear in the SEM photos of the coating surface. This phenomenon also explains the test results of adhesion strength, impact strength and flexibility.

QAS has a certain hydrophilicity, so the static water contact angles of the coatings show a downward trend with the increase of its addition amount, which is shown in Fig. 5. It can be seen from the test results that when the mass fraction of QAS exceeds 8 %, the static water contact angle of the coating drops below 100° , which can be considered that its surface performance is no longer qualified.

3.2. Anti-corrosion property

The corrosion protection of the primer is crucial in the coating package system. To explore the long-term corrosion protection performance and the corresponding electrochemical process, Nyquist, phase, and Bode plots of different coating specimens after 1, 5, 15, 25, 45, and 65 days immersion in 3.5 wt% NaCl solutions were recorded, as shown in Fig. 6.

At initial immersion, $|Z|_{0.01\text{Hz}}$ reflects the resistance of the coating pores to the penetration of corrosive electrolytes. The low-frequency impedance values of the EP and EP/QAS (3 wt%) coatings were $1.20 \times 10^{10}\ (\Omega\cdot\text{cm}^2)$ and $1.85 \times 10^{10}\ (\Omega\cdot\text{cm}^2)$, respectively. During the entire immersion process, the low-frequency impedance values of the EP and EP/QAS (3 wt%) coatings have remained stable at about $10^9\ (\Omega\cdot\text{cm}^2)$. After prolonged exposure, the impedance value changes very little, which confirms the stability of the coating in corrosive environments. After immersion for 65 days, the Nyquist diagram still showed a stable impedance value, the capacitive reactance arc radius did not decrease, and there was no diffusion tail. This indicates that the compactness of the EP/QAS (3 wt%) coating plays a barrier role and prevents corrosive media from diffusing into the substrate through the coating. Similarly, for the EP/QAS (5 wt%) coating, the initial impedance value is $9.76 \times 10^8\ (\Omega\cdot\text{cm}^2)$, which is slightly lower than the epoxy coating without QAS and the EP/QAS (3 wt%) coating. After immersing for 65 days, the resistance value of the coating is still stable at $10^7\ (\Omega\cdot\text{cm}^2)$. The capacitive reactance arc is slightly reduced, but there is no diffusion tail, which shows that the barrier properties of the coating did not lose quickly. Besides, the lower impedance value during the initial soaking can be explained by the following factors. First, increasing the content of

QAS increases the roughness of the coating, which can lead to uneven current distribution on the immersion surface. Secondly, excessive QAS particles might increase the stress in the matrix, leading to the formation of defects and voids in the coating and accelerating the corrosion rate. Therefore, during the entire immersion process, the EP/QAS (3 wt%) coating exhibits relatively efficient corrosion resistance.

The high-frequency phase difference was also an important parameter to evaluate the protective property of coating (Honarvar Nazari et al., n.d.; Haruyama and Sudo, 1993). For EP/QAS (1 wt%) coating, two-time constants appeared in the figure when the immersion time was 65d, and the impedance modulus significantly decreased at 0.01 Hz ($|Z|$, 0.01 Hz). It indicates that the corrosive medium has entered the coating, leading to corrosion of the metal surface. For EP/QAS (3 wt%) coating, the high-frequency phase Angle was 90.6° at the beginning of immersion. In the process of immersion of 3.5 % NaCl solutions, there was only one time constant in the figure, and the high-frequency phase angle kept stable at about 83° . From these results, it could be preliminarily determined that the phase angle value of the EP/QAS (3 wt%) coating is relatively high after immersion for 65 days, further explaining its long-term corrosion resistance performance during the whole immersion time.

To further analyze the protective performance of the coating, the equivalent circuit was analyzed as shown in Fig. 7. At the beginning of immersion; the R_c decreases gradually as the electrolyte solution penetrates the coating through micropores or defects in the coating. However, due to the thickness of the coating and the blocking effect of pigments and fillers, the R_c has a slight decrease, so the coating still has excellent anti-corrosion properties. This phenomenon can also be reflected in the Nyquist plots (Fig. 6). At this time, all coatings show a large impedance with only one time constant, and the data can be fitted with the model (a) in Fig. 7. In the middle of immersion, the R_c rapidly decreases by 1–2 orders of magnitude when the electrolyte penetrates the coating/metal interface, and the metal under the coating corrodes. During the subsequent immersion, R_c was maintained at a relatively stable value, indicating that electrolyte diffusion remained stable. In this stage, the value of R_c increases slightly and then remains stable. One possible cause was that the pores or defects in the coating were blocked by the corrosion products of the metal, resulting in the reduction of water and ion permeation channels. Model (b) in Fig. 7 can be used for the fitting procedure. In the later stage of immersion, the surface of the metal substrate corroded over a large area, and the R_c decreased rapidly. This phenomenon revealed that the electrolyte rapidly diffused into the coating, and the coating lost its protective effect. This period was controlled by the diffusion process, and the fitting process adopts the equivalent circuit model (c) in Fig. 7.

The NSS test was carried out to understand the corrosion resistance of the coating further. The photographs of all coated specimens and the

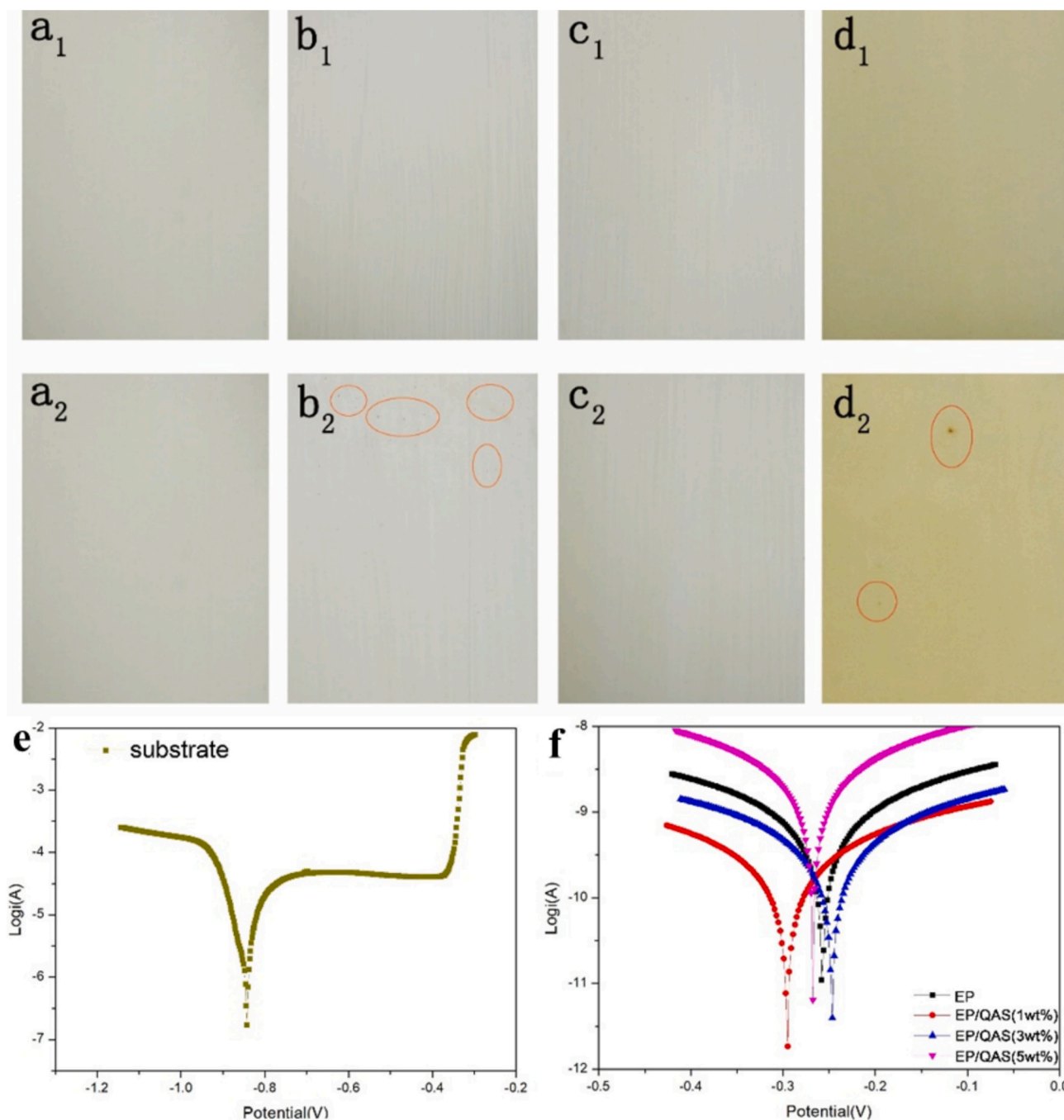


Fig. 8. Digital photographs of coating samples before and after 500 h immersion in 5 wt% NaCl solution: (a1) and (a2) EP, (b1) and (b2) EP/QAS (1 wt%), (c1) and (c2) EP/QAS (3 wt%), (d1) and (d2) EP/QAS (5 wt%). From top to bottom: before immersion and after 500 h immersion. Polarization curve with (e) substrate, (f) EP/QAS coatings measured in 3.5 wt% NaCl solution.

ones after 500 h immersion are shown in Fig. 8(a-d). It is easily found that after 500 h NSS tests, the EP/QAS (5 wt%) coating has visibly localized pitting corruptions (Fig. 8(d₂)), indicating the metal substrate has been seriously corroded. As can be seen from Fig. 8, the EP/QAS (5 wt%) coating was darker than that of other coatings. The reason might be that excessive mass fractions of QAS were added to EP/QAS (5 wt%) coating, resulting in agglomeration.

The Tafel polarization curve is used to further evaluate the anti-corrosion performance of the coating. Fig. 8(e-f) reveals the electrochemical polarization behavior of EP, EP/QAS (1 wt%), EP/QAS (3 wt %), and EP/QAS (5 wt%) coatings after 1 d of immersion in 3.5 wt% NaCl solution. The Tafel extrapolation method was used to obtain the

electrochemical corrosion parameters, including the corrosion current density (I_{corr}) and the corrosion potential (E_{corr}) from the polarization diagram, as summarized in Table S6. For the EP coating, the E_{corr} value is -0.26 V, the I_{corr} value is 8.43×10^{-11} A·cm⁻², the CR is 9.79×10^{-7} mm/year, and the R_p is 7.82×10^7 Ω. The E_{corr} value of EP/QAS (3 wt%) coating is -0.25 V, the I_{corr} value is 1.77×10^{-11} A·cm⁻², and the CR value is 2.05×10^{-7} mm/year. Therefore, EP/QAS (3 wt%) coating exhibits a significantly inhibitory effect on the corrosion of substrates compared to EP coating. With the further increase of the additional amount, the I_{corr} value and CR of the coating showed a trend of first increasing and then decreasing. This shows that adding an appropriate amount of QAS can improve the anti-corrosion performance of the

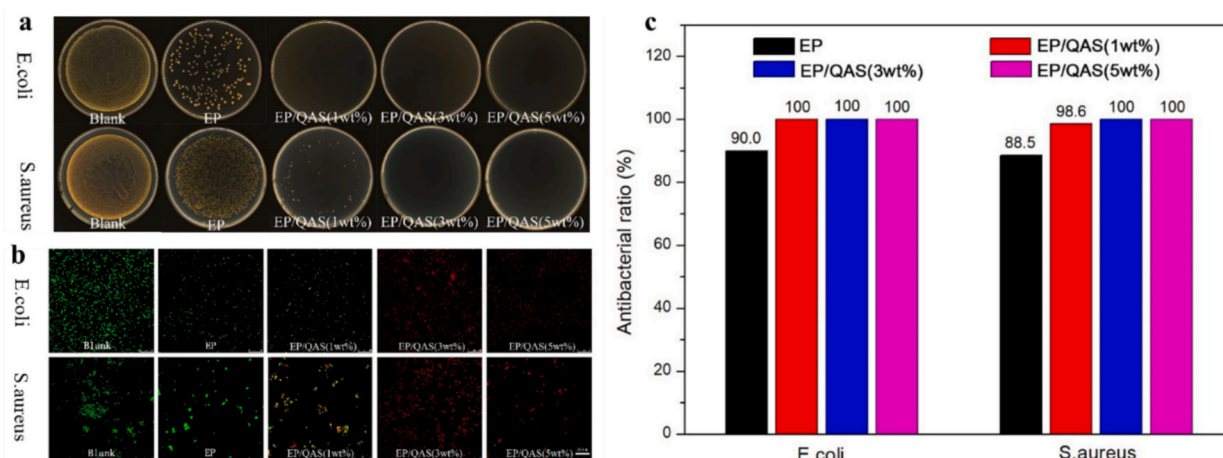


Fig. 9. (a) The Optical images of *E. coli* and *S. aureus* in flat colony counting method. (b) Fluorescence micrographs of *E. coli* and *S. aureus* stained on EP, EP/QAS (1 wt%), EP/QAS (3 wt%), and EP/QAS (5 wt%) coatings. (c) Anti-bacterial rate of EP, EP/QAS (1 wt%), EP/QAS (3 wt%), and EP/QAS (5 wt%). (The bacteria concentration was 1×10^5 CFU/mL).

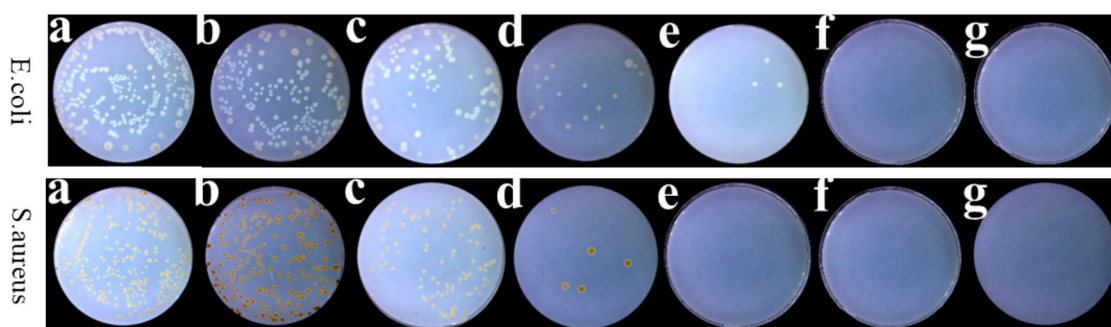


Fig. 10. Optical images of the coating added with different content of QAS interacting on *S. aureus*. The mass fraction of QAS is (a) blank control, (b) 0, (c) 2 %, (d) 4 %, (e) 6 %, (f) 8 %, (g) 10 %.

coating, but excessive addition of QAS might lead to a decrease in the anti-corrosion performance of the coating. In particular, the EP/QAS (3 wt%) coating showed the most significant corrosion resistance in all samples, and the corrosion rate was the lowest.

3.3. Anti-bacterial property

3.3.1. Antibacterial properties of the primer

To demonstrate the Antibacterial activity of coatings, *E. coli* and *S. aureus* were selected as representative gram-negative and gram-positive bacteria, respectively. Pristine and EP coating samples were prepared as controls, and the antibacterial properties of the coating were evaluated using the flat colony counting method.

As shown in Fig. 9(a and b), the growth and aggregation of bacteria on different coating surfaces. Multiple colonies were formed in the blank group and the pure EP coating after culture, while the number of various colonies decreased sharply after the QAS was added. The colony number of viable *S. aureus* on EP, EP/QAS (1 wt%), EP/QAS (3 wt%), and EP/QAS (5 wt%) coatings are 229, 27, 0, and 0 units, respectively. This indicates that the introduced QAS enables the coating to enhance its anti-bacterial properties.

The number of CFUs to quantify the growth of bacteria on surfaces was evaluated statistically in Fig. 9c. The anti-bacterial rate of *E. coli* inoculated on EP, EP/QAS (1 wt%), EP/QAS (3 wt%), and EP/QAS (5 wt%) coatings were 90.0 %, 100 %, 100 %, and 100 %, respectively, while the anti-bacterial rates of *S. aureus* were 88.5 %, 98.6 %, 100 %, and 100 %, respectively. It can be seen that after the introduction of QAS, the anti-bacterial rate of the coating on *E. coli* and *S. aureus* increased to

>98 %. EP/QAS (1 wt%) coating exhibits excellent resistance to *E. coli* with an inhibition rate reaching 100 %. For *S. aureus*, an inhibition ratio (98.6 %) was obtained by the same method. Therefore, the inhibitory effect of the coating on *E. coli* was slightly more significant than that on *S. aureus* (Buskens et al., 2012). Also, the reason why EP coatings have anti-bacterial effects might be that other components in the coating have anti-bacterial effects (Zhao et al., 2020).

Additionally, a Live/Dead assay was employed to further assess the anti-bacterial activity of coatings (Fig. 9b). After being incubated on the surface of the control or coatings, the bacteria were stained, and fluorescent images were taken. As seen in Fig. 9, strong green fluorescence was observed in the EP coating, indicating that the coating has strong bacterial activity. There is a significant increase in the number of inactivated, red-stained bacteria on the surface of coatings compared with that on the control surface. On the coating surface of EP/QAS (3 wt%) and EP/QAS (5 wt%) coatings especially, a large number of apoptotic bacteria and dead bacteria appeared. Besides, coatings exhibited stronger anti-bacterial activity against *E. coli* bacteria than *S. aureus* bacteria, which was in good agreement with the colony counting method results. The results show that the addition of QAS has a significant effect on the anti-bacterial performance of the coating. The greater the mass proportions were, the better the anti-bacterial activity was.

3.3.2. Anti-bacterial properties of the topcoat

Fig. 10 shows the anti-bacterial results of coatings with different QAS mass fractions against *E. coli* and *S. aureus*. This group of tests aims to show the anti-bacterial properties of other components except for QAS. It can be seen from the figures that QAS has prominent anti-bacterial

Table 2

Test results of the matching coatings in the ocean hanging test.

Sample	Before immersion	Immersed 2 weeks	Immersed 4 weeks	Immersed 8 weeks
A ₃				
B ₁				
B ₂				
B ₃				
B ₄				
B ₅				

properties against both bacteria. From the calculation results of the anti-bacterial test, the higher the dosage of QAS, the better the anti-bacterial effect performance. When the mass fraction of QAS reaches 6 % and 8 %, the coatings showed 100 % antimicrobial activity against *S. aureus* and *E. coli*. At the same time, the horizontal comparison shows that the anti-bacterial ability of QAS against *S. aureus* is more potent than that of *E. coli*. This is consistent with the results in the references (Gottenbos et al., 2003; Yin and Liu, 2019; Zhang et al., 2015b). The reason for this phenomenon is that the cell membrane structures of *E. coli* and *S. aureus* are different. *E. coli* and *S. aureus* used in the experiment are typical gram-negative bacteria and gram-positive bacteria respectively. The cell wall of gram-positive bacteria is mainly composed of peptidoglycans and acidic polysaccharides, while that of gram-negative bacteria is primarily composed of peptidoglycans and lipids (Rawlinson et al., 2010). In addition, gram-negative bacteria cells have an additional outer membrane outside the peptidoglycan, and the cell membrane is closely

connected with the cell wall. At the same time, the peptidoglycan structure of gram-positive bacteria cells is relatively loose, and the cell wall contains teichoic acid, which is not found in gram-negative bacteria. Part of the reason why the cell surface is negatively charged is that teichoic acid is negatively charged. Therefore, gram-positive bacteria are more sensitive to quaternary ammonium salt anti-bacterial agents. Based on the above reasons, the anti-bacterial effect of QAS against *E. coli* is better than that against *S. aureus* in this test.

3.4. Ocean hanging test

The antifouling performance of the matching coatings was investigated by the ocean hanging test in the East China Sea. Group A₃ was tested as a control. The photos of the coatings after immersion for 2, 4, and 8 weeks are shown in Table 2. The A₃ panel was covered by tiny fouling organisms such as diatom, and the overall appearance of the panel was tea-brown. QAS in the coatings significantly improved the anti-fouling performance of the coatings in static water. B₁ and B₂ coatings were only partly covered by fouling organisms after 8 weeks. There were still no apparent fouling organisms on the surface of the B₃, B₄, and B₅ coatings after soaking for 8 weeks.

3.5. Anticorrosion and anti-fouling mechanism

In the primer, QAS has enough cations to react with epoxy resin during the curing process, not only forming a tighter three-dimensional cross-linked structure but also producing good dispersion in the coating. The coating has a richer crosslinking density through the reaction with QAS in the EP/QAS (3 wt%) primer. The much richer crosslinking densities act as better physical barriers to prevent the electrolytes from passing through the coatings illustrated in Fig. 11.

However, when the doped QAS in the coating reaches 5 wt%, free QAS exists in the coating. These dissociated QAS can further form large particles due to the agglomeration, resulting in an increased molecular chain gap, which is conducive to the formation of electrolyte diffusion channels. The reasons can be accounted for by the poor corrosion protection for the EP/QAS coating with 5 wt%.

The synthesis mechanism of topcoat is essentially a reaction between hydroxyl and isocyanate groups. PPG, PDMS, and IPDI are all bifunctional group compounds, so the final polymer is a chained structure. At the same time, the existence of PPG and PDMS in the polymer gives the molecular chains a certain flexibility, which makes the physical entanglement between the molecular chains and finally forms a three-dimensional network structure. Powdered QAS is dispersed in the gap between the molecular chains like fillers, and the hydrogen bonds

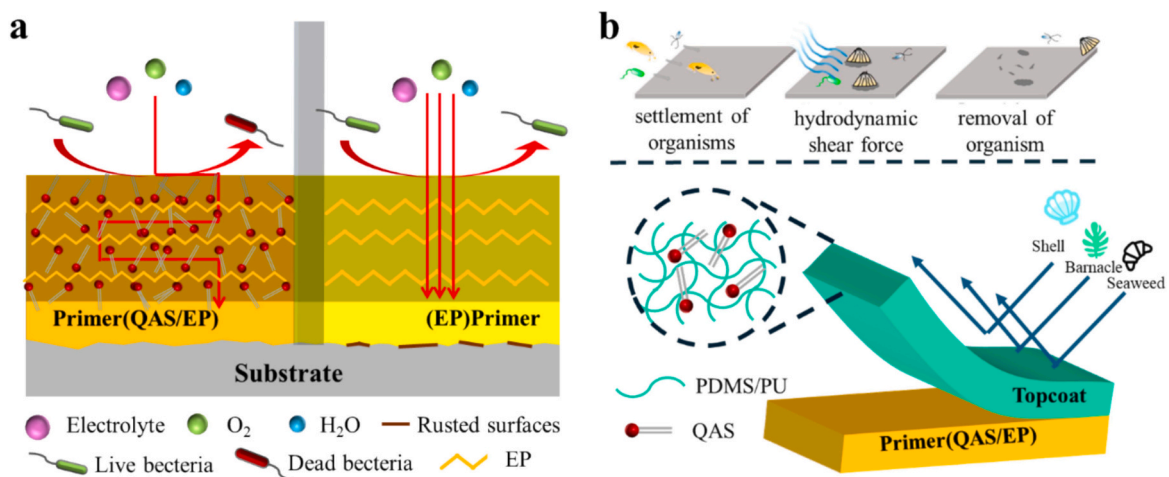


Fig. 11. (a) Protection mechanism diagram of the EP and EP/QAS coatings. (b) The antibacterial mechanism of QAS.

between them and the polymer molecular chains make the coating system more stable. The reasonable explanation for this phenomenon is that the hydrogen bonds between QAS and the molecular chains can not only form a tighter three-dimensional cross-linked structure but also produce good dispersion in the coatings. It should be noted that if the amount of QAS is excessive, the powder will be agglomerated, which might damage the continuity of the molecular chains and ultimately destroy the structure and performance of the coatings.

The best antifouling performance of the topcoat was achieved after doping with 6 wt% QAS, whereas less or excessive QAS deteriorated performance. QAS plays a bactericidal role in the coatings, so the antibacterial ability of the coatings is limited if its addition amount is insufficient. Nevertheless, when the doped QAS in the coatings exceeds 6 wt%, the dissociative QAS might further form large particles due to agglomeration. If the coatings are subjected to external force, these lumps will be damaged first due to the stress concentration, which will eventually lead to the decline of the mechanical properties of the coatings. At the same time, when the mass fraction of QAS reaches 6 %, the bactericidal rates of the coatings have reached or approached 100 %. The excellent bactericidal performance has significantly improved the comprehensive anti-fouling performance of the coatings.

4. Conclusions

In summary, a novel coating system was developed with quaternary ammonium salt (QAS) as the primary antibacterial agent, which demonstrated low toxicity, efficient anticorrosion and antifouling properties. The epoxy primer containing 3 % QAS exhibited the best mechanical properties, with a comprehensive antibacterial rate of 98 %. The results of EIS after 65d of seawater immersion and 500 h of neutral salt spray test confirmed that the primer had excellent corrosion prevention performance. Meanwhile, The best mechanical and surface properties of the topcoat were achieved when the molar ratio of PDMS and PPG was 1:2. A water contact angle of 114° was obtained, which effectively prevented the adhesion of fouling organisms. Additionally, the bactericidal rate of the topcoat against *E.coli* and *S. aureus* could exceed 99 % when the mass fraction of QAS was 6 %. Furthermore, the satisfactory results of the above coating system were confirmed by 8 weeks of hanging tests in the ocean, indicating potential prospects for long-term environmental protective application in marine engineering.

CRedit authorship contribution statement

Yihan Song: Writing – original draft, Validation, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Qi Wang:** Writing – review & editing, Validation, Software, Investigation, Formal analysis. **Yujie Ning:** Writing – review & editing, Investigation, Conceptualization. **Huayang Tian:** Writing – review & editing, Data curation, Conceptualization. **Bin Liu:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marpolbul.2024.117347>.

Data availability

The data that has been used is confidential.

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